



Review of artificial intelligence for enhancing intrusion detection in the internet of things

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ABSTRACT

Internet of Things is shaping the quality of living standard. With the rapid growth and expansion of adopting IoT-based approaches, their security represents a growing challenge for both manufacturers and consumers. There is a recent rising trend towards employing artificial intelligence approaches to enhance the security of IoT infrastructure. This survey paper focuses on reviewing recent developments in applying artificial intelligence to intrusion detection in the IoT domain. Selected articles are classified according to the applied AI algorithm. This study provides an in-depth survey highlighting the recent advances in artificial intelligence for improving the security of IoT. It summarizes and organizes the recent related research, then presents a comprehensive discussion on research challenges, open issues, and needed future research.

1. Introduction

Internet of Things (IoT) connects a multitude of smart devices including sensors, wearable devices, global positioning systems and mobile phones, etc. Its adoption in home automation, home security, video surveillance, connected goods, and tracking applications has resulted in phenomenal growth. According to Cisco Systems, machine to machine connections (M2M) will increase 2.4 times, from 6.1 billion in 2018 to 14.7 billion by 2023 (Cisco, 2018). Mobile M2M connections will increase fourfold from 1.2 billion in 2018 to 4.4 billion by 2023, representing a 30 percent CAGR (Cisco, 2018). IoT Analytics (IoT Analytics, 2023) showed an increasing growth of IoT devices compared to the non-IoT devices. It reported an annual increasing rate of 10% since 2018 and expected to reach 21.5 billion devices in 2025 exceeding three times the number of IoT devices of 2018 (IoT Analytics, 2023).

These devices can communicate and interconnect with each other to exchange data. IoT enables massive connectivity, reaching billions of connected devices communicating over the internet. With the rapid growth and widespread adoption of IoT, these devices can be found in homes, offices, transportation, healthcare, telecommunications, agriculture, and other settings. Most IoT devices have never been designed to handle security threats. Most of them are limited by resources, which makes them a potential attack surface for attackers. The vulnerability to malicious attacks is much higher in the IoT than in traditional networks due to its characteristics and use of communication protocols. Due to its

popularity, widespread use, mobility, and diversified distributed environment, IoT security represents a complex challenge that is completely distinct from other fields.

Over the past decade, artificial intelligence (AI) has seen tremendous success in several fields. Due to the unique nature of the IoT and the inability of traditional methods to meet its security needs, new security approaches are urgently needed. Several studies have presented various approaches for applying AI to the IoT. This paper examines the current approaches for using AI techniques to enhance the security of the IoT. Meanwhile, it conducts a comparative study of the current literature to analyze the potential and technical feasibility of AI in enhancing IoT security by discussing the challenges and exploring possibilities towards finding future research directions.

In summary, the following are the paper's main contributions:

1. Reviewing previous approaches of adopting AI for enhancing the IoT security.
2. Discussing the research gap and exploring the future research opportunities.

The remainder of this paper is structured as follows: Section 2 shows the materials and methods. Section 3 provides a context for some related concepts. Section 4 conducts a literature review of the previous related reviews. Section 5 highlights the motivation behind conducting this study. Section 6 investigates the potential of AI in improving IoT security

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through reviewing previously proposed approaches. Section 7 discusses the results and presents future research opportunities. Section 8 concludes the review.

2. Materials and methods

A number of studies have been presented for detecting network intrusions in IoT. To select relevant studies, a quantitative systematic review approach is used. This study is performed according to the criteria of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (MJ et al.). A thorough search was conducted using scientific electronic search engines on scientific databases including Web of Science, Scopus, IEEE Xplore, Science Direct, and Research Gate. The search is restricted to publications in scientific journals, conferences, and theses written in English. The search strategy consisted of all combinations of artificial intelligence (AI), machine learning (ML), deep learning (DL), intrusion detection, and IoT in the title, abstract, and keywords. Abstracts were screened and full-text papers obtained to identify primary research studies according to the following:

- The study has to be published in peer-reviewed journals, conferences, and/or workshops.
- All searches were conducted over the period from 2016 to 2023. The focus was only on published work during that period due to the fact that the trigger for this research field was the reported botnet malware (Mirai) in 2016 (De Donno et al., 2018) that disrupted the services of a major US Internet provider.
- The total study length should be at least four pages.

The following exclusion criteria were considered:

- The excluded study types are prefaces, editorials, summaries of tutorials, panels, and poster sessions.
- Studies that do not explicitly relate to IoT.
- Articles and papers providing unclear results or findings.

Two reviewers (M.S., M.M.) independently performed the searches and examined titles and abstracts to exclude irrelevant papers and produce a list of studies for full-text review in an iterative process. Any disagreement over inclusion or exclusion was discussed with the senior author (S.G.) and a consensus reached.

The literature search identified 2546 records in the initial database search. A PRISMA flow diagram for the systematic review is presented in Fig. 1. 108 studies met the inclusion criteria and were included in the review. 641 duplicated records were removed. 1905 records were screened to check the written language and 704 records were excluded as they are non-English written. Through comparing the titles, 351 papers were removed due to irrelevant titles. The next screening step was assessing the abstracts that resulted in excluding 482 papers with irrelevant abstracts. By applying the inclusion and exclusion prementioned criteria, 165 papers were excluded. Finally after through paper reading and through inspection, 108 studies were considered for the review.

Fig. 2 depicts the distribution of the selected papers over the years. It demonstrates an increasing research rate since the Mirai malware was reported up to the year 2019, as well as an exponential increase in the number of publications. The research is experiencing another hype after the year 2021 passing through the year 2022. This is accompanied by the implementation of new AI models. It is anticipated that this trend will continue in the coming years.

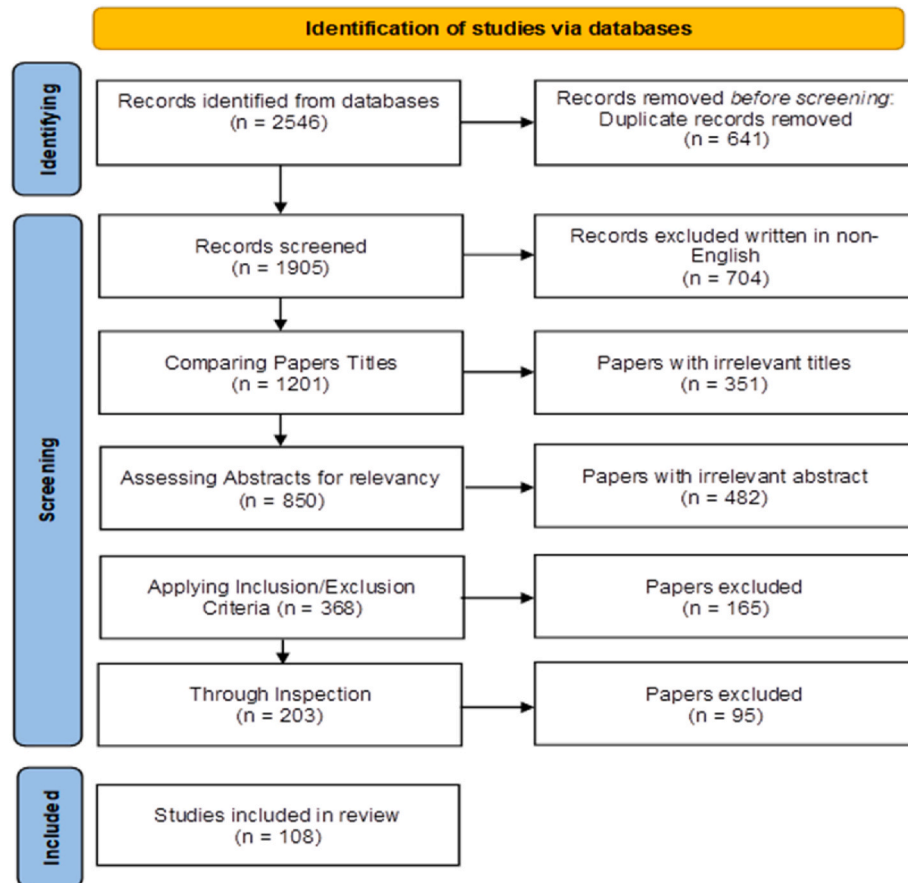


Fig. 1. PRISMA flow diagram of the study selection process.

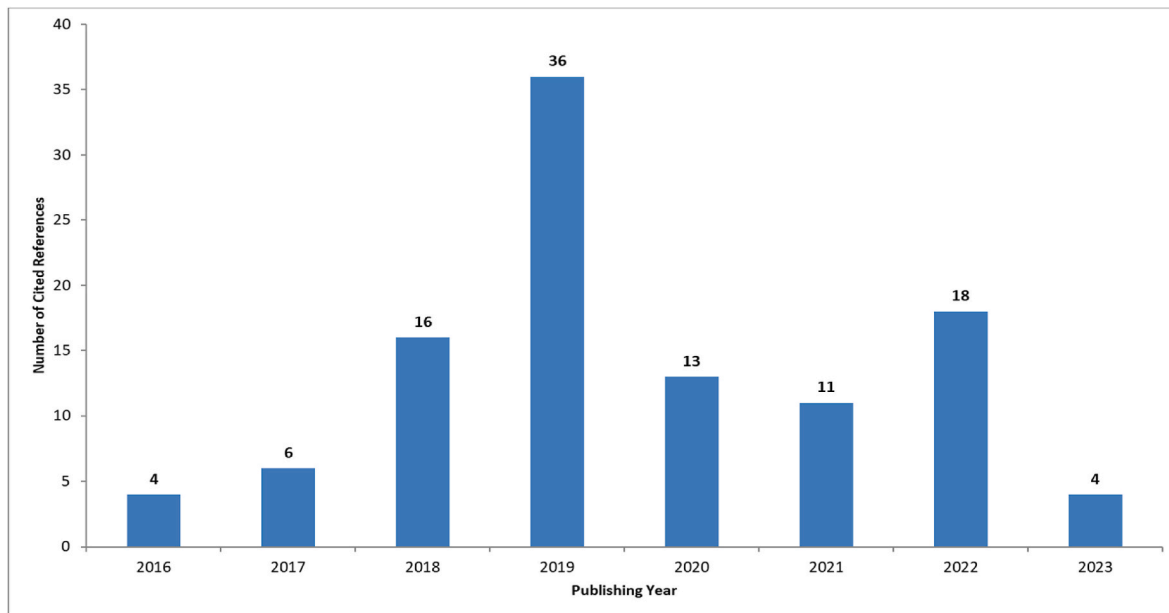


Fig. 2. Number of publications over years.

3. Background

This section explains briefly some of the key concepts and terms used in this paper to help the readers in understanding and better comprehending paper information.

3.1. Network Intrusion Detection Systems

Network Intrusion Detection System NIDS is defined as a system built to monitor and analyze network traffic to detect anomalies, intrusions, or violations (Azeez et al., 2020). The two primary methodologies for detecting intrusions into IoT systems are signature-based (a.k.a. misuse-based or knowledge-based) and anomaly based (a.k.a. behavior-based) approaches. A hybrid detection system is occasionally created by combining them, but it is difficult to implement and uses a lot of energy and resources (Dimolianis et al., 2021). While anomaly-based systems rely on traffic patterns to detect attacks, signature-based solutions classify traffic as malicious or benign based on known threats (Dimolianis et al., 2021). Signature-based systems work great against existing and well known attacks.

The time-consuming process of regularly updating the signature database is one of the issues with the signature-based method. Comparing the input to a large database becomes a computationally expensive and time-consuming process as the database size increases. Such a method is incapable of identifying zero-day attacks. An anomaly-based approach is the preferred approach because it examines regular traffic patterns and issues an alarm or restricts traffic anytime an anomalous traffic pattern is found. One advantage of employing anomaly-based systems is that they are effective at finding zero-day and unidentified assaults, although they could produce a lot of false-positive results (Radoglou-grammatikis and Sarigiannidis, 2018). Anomaly-based NIDS is one of the primary research areas focusing on its strong detection capabilities for large-scale, zero-day, and unidentified attacks.

3.2. NIDS classification

Intrusion detection and prevention systems (IDPS) can be divided into four categories based on how they are deployed and monitored. These categories are host-based (HIDPS), network-based (NIDPS), wireless-based (WIDPS), and network behavior analysis (NBA) (Azeez

et al., 2020).

The three basic categories for IDPS are: type of intruder, type of intrusion, detection technique (Azeez et al., 2020). While the detection techniques can be divided into six sub categories: misuse detection, anomaly-based detection, stateful protocol analysis, rule-based, supervised machine learning, and unsupervised machine learning (Azeez et al., 2020).

Depending on the training technique, anomaly-based NIDS can be divided into subcategories. These classifications are based on statistics, knowledge, and ML. Univariate, multivariate, and temporal series are all included in Statistical-based systems. Case-based, N-based, expert systems and descriptor languages are examples of finite state machines and rules used in knowledge-based systems.

A different method of categorization might be based on the CPU usage, throughput, and power of an NIDS. This classification applies primarily to contexts with limited resources environments, such as IoT.

4. Related work

For researchers, literature reviews are highly helpful in providing a summary of a particular research topic. They represent a road map for upcoming trends as well as state-of-the-art recommendations. This section discusses the previous reviews that are related to applying AI techniques for IoT intrusion detection. A number of reviews were presented for network security and IoT security; however, most of them lacked a review of the potential of AI in enhancing IoT security. According on the detection method, placement strategy, security threat, and validation approach, 18 papers about NIDS schemes in IoT were categorized by Zarpelo et al. in their classification survey (Zarpelo et al., 2017).

IoT security and privacy solutions were reviewed by Kouicem et al. in their report (Kouicem et al., 2018). They discussed how Blockchain could improve the security and privacy of the IoT. After reviewing the IoT model's security issues, Mohanta et al. predicted that ML, Blockchain, and AI might be used as potential solutions (Bhabendu Kumar Mohanta et al., 2020). The learning-based IoT security techniques, such as safe offloading, malware detection, access control, and authentication, were identified by Xiao et al. (Xiao et al., 2020). They discussed how to overcome the three issues of partial state observation, computing overhead and backup security measures in order to use ML in the IoT.

A study on deep learning (DL) methods utilized in IoT large data and

streaming was undertaken by [Mohammadi et al. \(2018\)](#). IoT security threats and attacks were covered by Al-Garadi et al. in their survey ([Al-garadi et al., 2020](#)). Their study examines the potential for ML/DL in IoT security through the three IoT layers, as well as broader IoT security risks. It does not have a focused scope on a certain layer. The potential of AI in IoT intrusion detection was not fully examined.

In the context of the aspects of IoT cyber security, [Ahmed and Askar \(2021\)](#) demonstrated some DL models, such as MLP, CNN, LSTM, and a hybrid model of CNN and LSTM. Their survey is dull, lacks the in-depth analysis, and concentrates on the definitions of the models.

IoT and ML were highlighted in a survey by da Costa et al. ([Kelton et al., 2019](#)) for numerous applied intelligent techniques for intrusion detection in computer networks. This work gives a thorough analysis of the possibility of integrating AI methods from the standpoint of improving intrusion detection in the IoT. [Ramos et al. \(2017\)](#) presented a survey that focused on models in contrast to other surveys. This paper covers the opportunities and limitations of various AI approaches for IoT security from the perspective of enhancing intrusions.

The approaches of intrusion detection systems in the IoT were categorized by [Tabassum et al. \(2019\)](#). Their attention was on hybrid and intelligent intrusion detection methods for the IoT. They discussed the layers of the IoT, communication protocols, security concerns associated with them, and the necessity of NIDS in both layered and protocol approaches. The majority of the intelligent NIDS approaches they surveyed are specialized for wireless sensor networks (WSN). [Santos et al. \(2018\)](#) presented a literature review on NIDS in the IoT field. They analyzed 20 related research works published between 2009 and 2017. They used a taxonomy based on characteristics like placement strategy, detection method, and security threat. They came to the conclusion that NIDS research in IoT is still in its infancy. They claimed that the reviewed works are incapable of detecting multiple attacks and do not cover the entire IoT domain.

[Ambarkar and Shekhar \(2020\)](#) examined six research publications that described the use of ML for IoT intrusion detection. Along with the detection rate, they discussed several ML based approaches. They demonstrated the performance of ML against Routing RPL Protocol attacks. However, the majority of those studies deal with regular computer networks rather than IoT environments.

An extensive survey of ML implementations in IoT was presented by [Cui et al. \(2018\)](#). They made an effort to address the main ML applications for the IoT and pertinent methodologies, such as IoT device identification, edge computing architecture, traffic profiling security, network management based on SDN-based IoT, and common IoT applications.

The potential of AI for IoT intrusion detection has been summarized in a number of articles, although much of the published research has not been reviewed.

5. Motivation

5.1. The special nature of internet of things

Compared to other networks, the IoT environment is different and has a unique nature ([Bertino and Islam, 2017](#)). The amount of digital data produced and transferred is enormous, with units ranging from Megabyte (10^6) and Brontobyte (10^{27}) of data to Geopbyte (10^{30}). IoT networks often generate large volumes of data, making it challenging to manually analyze and identify potential security threats.

IoT environment has a crosscutting nature through its multidisciplinary components that introduced new security challenges ([Al-garadi et al., 2020](#)). Those challenges result from the complexity and diversity of IoT devices. IoT has distinct properties such as limited resources, specific standards and protocols ([Zarpelão et al., 2017](#)). Its devices are mainly designed for functionality rather than security. For example, IoT-based smart waste management systems use sensors and other IoT devices to monitor waste levels, optimize waste collection routes, and

reduce overall waste management costs ([Hussain et al., 2020](#)). However, they also create new security challenges, such as the risk of unauthorized access to the IoT devices and the data they collect.

Moreover, most of these devices are working in an unattended environment, which adds a new security challenge to an already challenging environment. Consequently, applying traditional security protection mechanisms to IoT-connected devices is inadequate. Due to the diversity in IoT environments and different structures, selecting suitable security measure adds another complexity to an already complex and challenging process ([Bertino and Islam, 2017](#)). Some researchers consider Cyber security in enterprise distributed networks challenging; but securitization when adding IoT is a nightmare ([Benayas-ayuso et al., 2021](#)).

Shen et al. suggested in ([Shen et al., 2018](#)) a fog computing-based approach to identify and limit malware diffusion in order to get around the drawbacks of using NIDS on smart objects. An NIDS was used to build the suggested malware detection system, while cloud and fog computing were used for deployment. Due to restricted resources, other researchers have highlighted the need for an IoT middleware, such as da [Cruz et al. \(2018\)](#). Such middleware can be helpful for the implementation of AI-based decision-making systems. Additionally, Ferreira et al. presented UIoT ([Gabriel et al., 2017](#)), a set of security mechanisms for IoT middleware. Farhana et al. provided an overview of the primary challenges facing IoT in terms of privacy, security, and interoperability ([Javed et al., 2018](#)).

5.2. Intrusion detection techniques for the IoT

Information security researchers are encountering considerable hurdles when trying to design NIDS for IoT environments because of the unique characteristics of IoT that were covered in the previous section ([Zarpelão et al., 2017](#)). Traditional security software cannot be installed in IoT systems due to the lack of computing and storage capacity.

Due to the diverse nature, the size of typical activity, and the rise in vulnerabilities brought on by the exponential proliferation of IoT devices, traditional NIDS are ineffective in the IoT context. IoT privacy protection and intrusion detection are inherently difficult and complex tasks ([Kumar et al., 2017](#)).

5.3. IoT security

The ISO/IEC 30141 IoT reference architecture defines a three-layer architecture for the IoT environment: the terminal perception layer, the network transport layer, and the application service layer ([ISO/IEC 30141 INTERNATIONAL STANDARD, 2018](#)). Securing the IoT includes securing all those layers as well as the interfacing between them. This study focuses on the security of the network transport layer. The security perspectives in the application and terminal perception layers are out of the scope of this study.

The network layer is responsible for data transmission between the semantic perception layer and the application service layer. It uses non-cellular or long-distance networks (Bluetooth, ZigBee, Wi-Fi, etc.), cellular or short-distance networks (LoRa, eMTC, NB-IoT, etc.), in addition to the internet for authentication data encoding, and transmission. Many attacks are targeting the transport layer in an IoT environment, such as Denial of Service (DoS) and Distributed Denial of Service (DDoS). A famous attack is Mirai, launched 2016, which attacked more than 100,000 devices at that time ([De Donno et al., 2018](#)). DoS is an example of an active interruption attack. Mirai's source code was leaked in 2017, leading to development of new IoT malware. The leaked source code was used to develop variants of Mirai e.g. kiddies, Remaiten, Hajime, Reaper, Satori, Masuta, Linux.Darllloz, Amnesia and others.

Three methods of applying ML for identifying DDoS attacks in IoT were highlighted by [Wehbi et al. \(2019\)](#). It lacks the sophistication of the other IoT attacks. Their survey is limited to only three published studies

from 2018 and can be considered complementary to the previous reviews. However, although the development of IoT has reached maturity, no unified IoT security standards have yet been created (Al-garadi et al., 2020).

6. AI potential for IoT NIDS

Traditional security approaches have been effective in protecting networks and systems from known threats. However, they often fall short in detecting and responding to emerging threats and zero-day attacks, which are becoming increasingly common in IoT systems. Additionally, traditional security approaches can be resource-intensive and may not be well-suited for the dynamic and complex nature of IoT systems.

AI technologies have shown significant progress in improving many fields. But to what extent AI techniques can enhance the IoT systems security. AI-based intrusion detection systems have the potential to address many of the limitations of traditional security approaches. Through analyzing large amounts of data in real-time, AI-based intrusion detection systems can identify and respond to emerging threats quickly and accurately. Additionally, AI-based intrusion detection systems can adapt to changing environments and learn from experience, making them well-suited for the dynamic and complex nature of IoT systems.

By leveraging the potential of AI, intrusion detection systems can fill the gaps left by traditional security approaches and provide more comprehensive and effective protection for IoT environments. AI can contribute in enhancing IoT security through providing its three main capabilities. The first capability is through teaching the models through a prior knowledge or labeled data which is known by supervised learning approach. The second capability is extracting the hidden knowledge without prior labeling through an unsupervised learning approach. The third capability is through gradual model by using the reward and punishment technique which is known by reinforcement learning. AI can put the NIDS in IoT into a new dimension.

For example, AI-based intrusion detection systems can analyze network traffic patterns and identify anomalies that may indicate suspicious activity, even if the activity has not been seen before. AI-based intrusion detection systems can also learn from their past experiences and adjust their behavior based on new data, allowing them to continually improve their accuracy and effectiveness over time. By learning from past instances, AI is able to forecast unknown future attacks. AI techniques make it possible to handle anomaly-based intrusion detection by examining deviations from historically normal traffic (Zarca et al., 2019).

In conclusion, AI has the potential to revolutionize the field of IoT intrusion detection by providing more comprehensive and effective protection against emerging threats and zero-day attacks. While there are still challenges to be addressed, continued research and development in this area will be crucial to realizing the full potential of AI in IoT intrusion detection and ensuring the security and privacy of IoT devices and networks.

6.1. Machine learning

6.1.1. Naïve Bayes

To classify IoT network traffic, Guezzaz et al. developed a hybrid technique based on Snort NIDS and Naïve Bayes classifier (NB) (Guezzaz et al., 2022).

Based on misuse detection, Snort NIDS was employed as the first layer to handle first intrusion. The second detection layer was created using a Naïve Bayes binary classifier. They used a portion of the KDD CUP 99 (Al Tobí and Duncan, 2018) dataset and outperformed Support Vector Machine (SVM) and k Nearest Neighbors (kNN) in terms of accuracy, achieving a competitive rate of 96.45% compared to SVM 93% and kNN 94.4%.

6.1.2. Support vector machine

Despite its long learning time, SVM is also adopted in many approaches. SVM is better suited for limited datasets because it takes a while to find a hyper plane. An approach to multi-level hybrid intrusion detection was presented by Al-Yaseen et al. SVM and an extreme learning machine are used (Al-yaseen et al., 2016). To enhance the performance of classifiers, a high-quality training dataset is created using a modified K-means technique. They attained an accuracy of 95.75% using the KDD Cup 99 dataset (Al Tobí and Duncan, 2018).

Researchers have been looking into the prospect of using AI approaches in DDoS attack detection due to the growing threat of DDoS attacks in the IoT. Bakker et al. investigated the implementation of ML algorithms with traffic classification architecture to identify DDoS attacks due to SDN's ability to collect data from the network into a single location (Bakker et al., 2018). They employed three classifiers in an offline environment. Then, a physical network test bed was employed to recreate a DDoS attack scenario for evaluation. Seven ML techniques are chosen: Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), SVM, kNN, NB, Decision Trees (DT), and Random Forest (RF). They trained their model using data from the Information Security Center of Excellence (ISCX) dataset (Information Security Center of Excellence (ISCX)). For all classifiers, the detection accuracy rates ranged between 92% and 93%. SVM was shown to have the maximum accuracy when classification was enabled by the Data Plane Auxiliary Engine (DPAE) (Hayes et al., 2017). Such approaches focus only on one type of attack i.e. DDoS.

For forensics reasons, Koroniotis et al. presented a brand-new dataset named Bot-IoT (Koroniotis et al., 2019). It was made using a simulated IoT scenario with different attack kinds. Smart refrigerators, smart garage doors, weather monitoring systems, smart lights, and smart thermostats were examples of IoT devices. They used the SVM, RNN, and LSTM-RNN ML algorithms to evaluate this dataset. Their respective detection accuracy percentages were 99.98%, 97.90%, and 98.05%. The SVM based technique had the highest reported accuracy.

In order to identify anomalies in the IoT, Ioannou and Vassiliou (2019) presented a c-support vector machine (c-SVM) based technique. When there are multiple classes of data, c-SVM is a type of SVM that can be used. With a similar network topology, the greatest accuracy of 100% was attained. But when the model was trained using various topologies, the accuracy dropped to 81%.

An SVM-based technique was created by Jan et al. (2019) to identify an adversary trying to introduce extraneous data into the IoT network. They trained and evaluated their models using the CICNIDS2017 dataset (Panigrahi and Borah, 2018). Their method had a 98.03% detection accuracy rate. They compared the performance of the SVM-based classifier with that of the NN, k-NN, and DT ML-based classifiers, and found that its accuracy is the highest.

A method for identifying botnets based on behavioral analysis was presented by Goyal et al. (2019). They tested DT, SVM, Artificial Neural Networks (ANN), and Logistic Regression (LR). While no accuracy was reported for DT, the reported accuracy rates were 99.23%, 99.86%, and 99.74%, respectively. The SVM algorithm produced the highest accuracy.

SVM was utilized by Bagaa et al. (2020) to create an NIDS for the IoT. The detection of anomalies was 99.71% accurate. In a test bed for smart buildings, they produced a real dataset using real sensor data from four separate rooms. The dataset contains measurements of 67,876 samples.

Jingyu et al. (Liu et al., 2021) proposed an intrusion detection approach for IoT such that a particle swarm optimization-based gradient descent is used for features extraction then one class SVM (OC-SVM) is used for classification. They employed UNSW-NB15 dataset (Moustafa and Slay, 2015) for evaluation and the reported detection accuracy is 86.68%.

6.1.3. K nearest neighbor

Kumar and Lim proposed in (Kumar and Lim, 2019) a modular

intrusion detection solution called Early Detection of IoT Malware Activity (EDIMA). The five modules that make up EDIMA's modular architecture are the ML Classifier, ML Model Constructor, Packet Traffic Feature Database, Policy Module, and an optional sub-sampling module. The Gaussian Naive Bayes, kNN, and RF classification methods were all examined by the authors. The percentages of accuracy were 88.8%, 94.44%, and 77.78%, respectively. The maximum detection accuracy is reportedly attained by the kNN. An ML classifier module is deployed on the access gateway to gather samples of incoming traffic, extract feature vectors, and then classify them using an ML model created by an ML model constructor. A supervised classification technique is used by the ML Model Constructor to categorize data. The model will be retrained and compared to the existing model whenever new malware is detected. Otherwise, there won't be any change. The feature vectors are kept in the packet traffic feature database. To determine the appropriate course of action, the policy module comprises defined policies. For larger IoT environments, the sub-sampling module is an optional component.

A layered approach for IoT intrusion detection was suggested by Pajouh et al. (2019). Their approach is made up of two engines, one for dimension reduction and one for classification. The dimension reduction engine is two-tier engine that consists of component analysis based on NB classifier to classify attack records. Then, the refinement stage of the second filtering stage uses linear discriminate analysis based on the Certainty Factor version of K-nearest neighbor (CF-kNN). Their method was dubbed Two-layer Dimension Reduction and Two-Tier Classification (TDTC). They were able to achieve an 84.86% detection accuracy using the NSL-KDD dataset (NSL-KDD dataset).

Alsamiri and Alsubhi (2019) evaluated seven ML algorithms in terms of detecting IoT network attacks. They used Bot-IoT (Koroniotis et al., 2019) for evaluation and implementation. The reported highest detection accuracy rate was 99% for kNN. Lower performance of 97% was reported for RF, Iterative Dichotomiser 3 (ID3), and Adaboost. QDA, MLP, and NB achieved unsatisfactory accuracy rates of 87%, 84%, and 79%, respectively.

6.1.4. Decision Trees

Feature selection was used by Bahsi et al. (2018) to reduce the number of features and increase accuracy. They evaluated two ML methods for classification i.e., DT and kNN. They claimed that DT's classification accuracy was 98.97%, whereas kNN's was 94.97%. Nine IoT devices, including a security camera, webcam, baby monitor, thermostat, and doorbell, were used to form a simulation IoT network. Three labels normal, Bashlite (also known as Gafgyt), and Mirai are included in the dataset. There are 502,605 normal, 2,835,317 Bashlite, and 2,935,131 Mirai records in the dataset packets.

Deep belief and decision tree ML were used by Aloqaily et al. (2019) for intrusion detection in the context of connected automobiles. The proposed model had a 99.43% accuracy rate. 22,544 record simulation dataset was used. A supervised method with three layers was put forth by Anthi et al. (2019) for identifying and classifying IoT intrusions. Create a profile for each IoT device's typical behavior is one of the system's three key tasks. In the event of an attack, it locates malicious packets on the network and then classifies the attack's nature. They created a test bed for smart homes with 8 IoT devices. Four basic categories of attacks DoS, man in the middle (MITM), reconnaissance, and replay were injected into a total of 12 attacks. Nine classifiers were chosen to categorize unobserved data. Classifiers were chosen based on their classification speed, capacity for multiclass classification, and large feature space. NB, Bayesian Network, J48, Zero R, One R, Simple Logistic, SVM, Multi-Layer Perceptron (MLP), and RF make up the nine classifiers. The outcomes demonstrated that the DT J48 model had the best overall performance. Device profiling, wireless attack detection, and attack type classification all had reported ratings of 96.2%, 90.0%, and 98.0%, respectively.

Illy et al. combined different ML algorithms to build an ensemble classifier (Illy et al., 2019). They adopted the Decision Tree Bagging

Ensemble technique DT (BE). The NSL-KDD dataset (NSL-KDD dataset) was employed for training and evaluating models. For binary classification and attack classification, they attained accuracy of 85.81% and 84.25%, respectively.

A deep belief model for intrusion detection in smart medical environments was proposed by Manimurugan et al. (2020). The CICNIDS 2017 dataset was utilized (Panigrahi and Borah, 2018). Their model achieved good accuracy for benign class 99.37%; but unsatisfactory accuracy for anomalies. The highest detection accuracy was for a web attack at 98.37%. Brute Force and Port Scan were both detected at 97.71%. The lowest accuracy rates were 96.67% for DoS/DDoS and 96.37% for Infiltration.

6.1.5. Random Forest

Doshi et al. (2017) generated a labeled training dataset by simulating a local network of consumer IoT devices, which included both benign and malicious traffic. They tested five different ML classifiers on this generated and labeled dataset, including KNN, LSVM, DT, and RF using Gini impurity scores. According to their findings, combining stateful with stateless features improves accuracy. For all features, RF outperformed both KNN and DT with 98.9% and 99.5%, respectively, while LSVM performed the worst with 92.1% accuracy.

A DNS-based profiling system was offered by Dwyer et al. (2018) to identify Mirai-like botnet activity. Their strategy relies on the information returned by DNS requests and makes use of a RF classifier. They tested their strategy using actual honeypot datasets and contrasted it with kNN and Bayesian-based classifiers. The RF classifier got 99% accuracy, which was the highest. The highest accuracy was achieved by RF with 99%.

A ML-based approach for identifying DDoS attacks that works in two phases: detection and mitigation, was presented by Chaudhary and Gupta (2019). They created a dataset by capturing the traffic of an IoT environment comprised of PCs and Raspberry PI with Wireshark. The dataset includes 114,565 packets, of which 10,061 are benign. For classification, they evaluated four algorithms: RF, SVM, LR, and DT. The reported accuracy rates are 99.17%, 98.06%, 97.50%, and 98.34%, respectively.

Hasan et al. (2019) carried out comparative implementation research to determine the optimum algorithm for IoT intrusion detection and classification. They took into account ANN, DT, RF, LR, and SVM. They used the Pahl open source dataset, which contains fictitious data from the Internet of Things ecosystem known as the Distributed Smart Space Orchestration System (DS2OS). They stated that RF had the highest accuracy performance at 99.4%. Their research, however, did not address large datasets or other unknown problems.

An NIDS for IoT in a smart city based on RF algorithm was proposed by Alrashdi et al. (2019). The UNSW-NB15 (Manimurugan et al., 2020) dataset was utilized to evaluate their model (Moustafa and Slay, 2015). They reported 99.34% detection accuracy and the lowest false positive rate.

Thamilarasu et al. demonstrated a mobile agent based approach for detecting intrusions in the medical IoT (Thamilarasu et al., 2020). They simulated the topology of a hospital network on the Internet of Medical Things. They trained five supervised ML algorithms, i.e., SVM, DT, NB, KNN, and RF. They reported that the KNN and NB algorithms performed inappropriately. SVM, DT, and RF, on the other hand, did well. RF demonstrated the best performance, achieving a nearly 100% accuracy rate.

Hammoudeh and Aljaberi (2021) proposed intrusion detection system for IoT based on the gated recurrent unit (GRU) deep learning algorithm with flower pollination algorithm (FPA) for feature selection with an accuracy of 99.59%. They conducted an extensive experimental analysis for evaluating their approach against some ML based models i.e. DT, RF, LR, and an ensemble of several ML algorithms (SVM + DT + RF + LR + GDB). The accuracy rates were reported as 91.04, 89.39, 90.37, and 92.03, respectively. Their study was employed on KDD Cup 99

dataset (Al Tobi and Duncan, 2018).

Eskandari et al. presented an intelligent NIDS for the IoT utilizing lean ML techniques for one-class classification (Eskandari et al., 2020). They referred to their strategy as “Passban”. It was made up of iForest and LOF, two one-class classification methods. To replicate a typical smart home automation environment, they created an IoT test bed. They looked at two scenarios: distinct standalone NIDS and NIDS deployed directly on the IoT gateway. The four attacks they tested it against were HTTP brute force, port scanning, SYN flood, and an Secure Shell (SSH) brute force attack. The claimed detection precision ranged from 79% to 99%.

Saied and Guirguis conducted an evaluative study for evaluating the performance of tree based approaches for intrusion detection in IoT (Saied and Guirguis, 2023). They utilized the N-BaIoT dataset (Meidan et al., 2018) for models training and testing. They compared six different algorithms for binary detection (DT, RF, Bagging Meta Classifier (BMC), Adaptive Boost model (ADB), Gradient Descent Boosting (GDB), and Extreme Gradient Boosting (XGB). They reported the best performance using RF based model with a detection accuracy rate of 99.99%.

6.1.6. Extreme Gradient Boosting

Peer-to-peer P2P bots were identified by Kumar et al. using a two-step procedure that included detection and analysis (Kumar et al., 2019). XGB, DT, and RF are employed in the classification step with 10-fold cross validation. Their method had a 99.88% detection rate. They used traffic from the Waledac, Vinchuca, and Zeus botnets to train the peer-to-peer (P2P) botnet detection algorithm.

Eleven ML techniques for detecting intrusions in Contiki-NG-Based IoT Networks were examined by Liu et al. (2020). They claimed that, utilizing the NSL-KDD dataset, XGB had the greatest performance, with 97% accuracy.

Another strategy for identifying intrusions in the IoT was presented by Alqahtani et al. (2020) and is based on the XGB. Through fewer features in the N-BaIoT dataset (Naveed et al., 2020) and a reported accuracy of 99.97% for multi-class classification.

Using the log data produced by HVAC (heating, ventilation, and air conditioning) attacks; Khan et al. (2022) suggested a proactive interpretable prediction model to detect various sorts of security attacks. DT, RF, GDB, ADB, Light Gradient Boosting (LGB), XGB, and Categorical boosting were among the ML techniques utilized (CAB). They claimed that the best result, with 99.98% accuracy, was produced by the XGB classifier. The HVAC systems dataset from Elnour et al. was used in their investigation (Elnour et al., 2021).

A federated intrusion detection system was presented by Ashraf et al. (2022) for Blockchain-enabled IoT healthcare applications. Their strategy is based on federated learning with lightweight ANN. Additionally, it offers a distributed ledger for the aggregation of local weights using Blockchain technology. Blockchain is utilized for weights broadcasting after averaging the updated global weights. They utilized the BoT-IoT dataset to compare the use of ANN and XGB models. They compared using of ANN and XGB models through BoT-IoT dataset. The results indicate that ANN performs better than XGB with 99.99 than 98.96.

Alissa et al. (2022) proposed an DT, an XGB, and a LR models. They used UNSW-NB15 dataset in conjunction with the features correlation technique, which resulted in the removal of nine features. They reported that DT outperformed with 94% test accuracy with slightly higher accuracy than XGB, while LR achieved the worst accuracy.

The effectiveness of boosting approaches and non-boosting ensemble-based strategies was examined by Garg et al. (2022). They identified two types of attacks: IoT attacks and DDoS attacks, as binary class and multiclass output, respectively. Three datasets were used for the evaluation: BoT-IoT, IoT-23 and CIC-DDoS-2019. The XGB and LGB boosting techniques were also employed. LGB outperformed the competition with an accuracy of 94.79%.

Awotunde et al. (2023) proposed a boosting based model for intrusion detection in industrial internet of things networks. They

investigated the detection of various ensemble classifiers, such as XGB, Bagging, extra trees (ET), RF, and ADB. They utilized the Telemetry data of the TON_IoT datasets (ALSAEDI et al., 2020). The results indicated that the XGB showed the highest accuracy in detecting and classifying IIoT attacks.

Rani et al. (2023) compared several algorithms for intrusion detection in IoT environments i.e. LR, RF, XGB, and LGB. They utilized DS2OS dataset (DS2OS traffic traces, 2018) and reported that XGB and LGBM achieved almost equal accuracy of 99.92%.

6.1.7. Light Gradient Boosting

An LGB-based model for anomaly identification in an IoT environment was proposed by Bhoi et al. (B. B et al., 2022). They used gravitational search-based optimization (GSO) for optimizing LGB hyper parameters and compared it with particle swarm optimization (PSO). They used a simulated IoT sensors dataset called the IoT dataset that is cited in (François-Xavier Aubet, 2018). They reported an optimal accuracy of 100%.

6.1.8. Adaptive gradient boosting

A multi-class Adaptive Boost model (ADB) was suggested by Dash et al. (Dash and Rao, 2020) for forecasting the anomaly kind. For the model evaluation, they used the IoT security dataset from Kaggle DS2OS (DS2OS traffic traces, 2018). This dataset includes eight different kinds of anomalies: data probing, DoS attacks, malicious control, malicious operation, scan, espionage, and incorrect setup. They claimed a 95% accuracy rate for anomaly detection.

A hybrid technique based on ML and feature selection was suggested by Krishna et al. Krishna et al. (Krishna and Thangavelu, 2021). Recursive feature elimination is used to apply feature extraction to the NSL-KDD (NSL-KDD dataset) and NBaIoT datasets (Krishna and Thangavelu, 2021). They reported an accuracy of 99.98%. They compared it with the gradient boosting classifier, which achieved an accuracy of 99.30%.

Qasem et al. (Al-Hajja et al., 2021) employed ADB algorithm combined with DT for detecting cyber-attacks in IoT networks. They evaluated their model on the TON_IoT dataset (ALSAEDI et al., 2020) and reported an overall accuracy of 99.70 for multiclass classification of nine types of attacks: injection, password, ransomware, backdoor, scanning, MITM, DoS, DDoS, and cross-site scripting (XSS).

Hazman et al. (2022) proposed an approach for intrusion detection in IoT-based smart environments with Ensemble Learning. Furthermore, they employ ADB while combining various feature selection techniques such as Boruta, mutual information, and correlation. They tested their method on IoT-23, BoT-IoT (Koroniotis et al., 2019), and Edge-IIoT datasets with accuracy of 99.90%.

Al-Hajja et al. (Al-hajja and Al-Dala'ien, 2022) adopted an ensemble learning approach to create a botnet attack detection model. Their approach is to apply the voting-based probability to ensemble the three ML classifiers, i.e. ADB, random under sampling boosting model (RUS), and the bagged model. Individual performance for the classifiers chosen was 97.30%, 97.70%, and 96.20%, respectively. The proposed ensemble model performed at a rate of 99.60%.

6.2. Deep learning

The non-linear combination of features from the original data set can be learned through deep learning (DL). To minimize the complexity of model pre-processing time, it can be integrated with ML models. This section presents the DL based intrusion detection approaches in IoT.

6.2.1. Supervised deep learning

6.2.1.1. Multi-Layer Perceptron. For the purpose of identifying DoS and DDoS attacks, Hodo et al. (2016) proposed a MLP supervised neural

network. To train the model, a simulated IoT network simulation was used. It had five nodes of sensors in total. 1,184 benign and 2,121 malicious records were present in the collection. Their method has a 99.4% accuracy rate.

6.2.1.2. Artificial neural network. Cañedo and Skjellum (2016) examined the use of ANN to detect anomalies in the IoT. Using their own test bench, they built a real dataset. Temperature sensors served as the test bed's edge devices, while a Raspberry Pi served as the test bed's IoT gateway. They gathered 4,000 samples for their dataset. Although their model had a high accuracy of 99%, the IoT gateway's inadequate resources for real-time applications limited its ability.

Chang and Hsieh used the Apache Spark framework with a back propagation ANN-based detection algorithm (Chang and Hsieh, 2018). They combined the DARPA 1999 dataset (Anjum et al., 2021) with the CAIDA DDoS 2007 dataset for training the model. The real-time detection system acquired a detection rate of 87.18%, while the model attained a detection rate as high as 99.80%.

Mohamed et al. proposed a NN and RF based NIDS approach (Mohamed et al., 2018). They used RF to detect intrusions and applied NN to classify them. They used the UNSW-NB15 dataset (Moustafa and Slay, 2015) and achieved an average accuracy of 89%. They attempted to improve detection performance by changing the neural networks architecture, such as by using different residual layers and bypassing layers, but there was no improvement.

The bare minimum ANN configuration was utilized by Soe et al. to identify DDoS in the IoT (Soe et al., 2019a). Their simple approach consists of only one single hidden layer and one single output node. They employed the Bot-IoT dataset (Koroniotis et al., 2019) and adopted the synthetic minority over-sampling technique for data re-sampling to address the issue of data imbalance (small amount of benign samples compared to malicious traffic data). They claimed that 100% detection accuracy was achieved.

An ANN-based technique for IoT intrusion detection was presented by Fatayer and Azara (2019). They used learning vector quantization (LVQ), radial basis function (RBF), and multilayer perceptron (MLP) as their three ANN approaches. The KDD CUP 99 dataset (Al Tobi and Duncan, 2018) was utilized by them to assess their model. In comparison to the other versions, they claimed that LVQ2.1 had the best classification accuracy at 97.44%. Compared to LVQ2.1 and RBF, MLP achieved the maximum accuracy of 99.86%. An ANN-based method for identifying rogue IoT nodes was described by Khatun et al. (2019). The feed forward NN and back propagation training algorithm make up the model. It has four concealed layers, all of which make use of the sigmoid function. Their model was put up against NB, SVM, DT, and RF, but ANN came out on top with a detection accuracy of 77.51%.

An enhanced multiple input multiple output model for heterogeneous WSN was presented by Lakshmi and Prashant (Lakshmi and Prashanth, 2022). Their model focuses on the identification and prevention of Black hole attacks. For the detection, they used the Back Propagation Neural Network (BPNN) technique.

Soe et al. (2019b) proposed NIDS for detecting multiple specific attacks in a sequential manner. Each kind of attack is detected using a designated classifier. They adopted NN architecture. They used the N-BaIoT dataset (Meidan et al., 2018) for training NN models and for evaluation. Each model was configured to use the sigmoid activation function at the output layer, as they reported it was the best based on a comparative study they did. Each model was able to achieve 100% detection accuracy for detecting its designated attack.

6.2.1.3. Auto encoder. Deep auto encoders were used in IoT networks to find anomalies. Deep auto encoders were utilized by Meidan et al. (2018) to create the N-BaIoT network-based anomaly detection approach for the Internet of Things. Four hidden layers of encoders, each with a decreasing size of 75, 50, 33, and 25% of the input layer's

dimension, make up their method. The following layers were decoders with the same sizes as the encoders, but with a higher order beginning at 33 percent.

They made use of a dataset of actual traffic data that was collected from nine commercial IoT devices that had actually been attacked by Bashlite and Mirai. They trained and refined a deep auto encoder on 2/3 of the benign data for each of the 9 IoT devices. Each device's test data was made up of all the malicious data as well as the remaining third of benign data. They trained three other algorithms local outlier factor (LOF), one-class SVM, and isolation forest and tuned their hyper parameters in preparation for evaluation. They claimed that their strategy produced a TPR of 100%. It indicates that they were successful in identifying each and every attack.

6.2.1.4. Conventional neural network. Kan et al. (2021) employed Adaptive Particle Swarm Optimization Convolutional Neural Network (APSO-CNN) for intrusion detection in the IoT. APSO is a kind of evolutionary algorithm proposed by Kennedy and Eberhart and is inspired by the predatory behavior of birds. The particle location parameters were set up to correspond to each layer structure's variable hyper parameters. Under CNN's initial training phase, the cross-entropy loss function value was configured as the fitness value of the APSO. The speed and position of the particle swarm are updated repeatedly as a means of searching for the fitness value. The appropriate CNN structure parameters are acquired while adaptively changing the inertia weight component with the fitness value. The method was compared to SVM, FNN, and R-CNN, which had accuracy rates of 91%, 88%, and 89%, respectively, while APSO-CNN had an accuracy rate of 96%. A deep learning CNN-based technique for detecting intrusions in the IoT was proposed by Idrissi et al. (2021). They compared RNN, LSTM, and GRU to other deep learning algorithms using the Bot-IoT dataset (Koroniotis et al., 2019) for evaluation. They claimed that CNN had the best prediction execution time of less than 0.34 ms and the greatest detection accuracy rate of 99.94%.

Fatani et al. proposed a feature extraction method using CNNs to extract relevant features (Fatani et al., 2021). Also, they created a technique for feature selection by modifying the transient search optimization TSO algorithm and applying differential evolution DE operators. They abbreviated it as the TSO-DE algorithm. The purpose of DE is to better balance the phases of exploitation and exploration. They utilized four open datasets for evaluation: KDD Cup 99 (Al Tobi and Duncan, 2018), NSL-KDD (NSL-KDD dataset), Bot-IoT (Koroniotis et al., 2019), and CICNIDS 2017 (Panigrahi and Borah, 2018). For binary classification and 99.61% for multi classification, their method produced 99.62% detection accuracy.

Zhou et al. (Jingjing et al., 2022) proposed an intrusion detection model for WSN based on CNN and GRU. They employed WSN-DS-2016 dataset (I et al., 2016) which includes black hole, gray hole, flooding, scheduling attacks. The reported accuracy was 99.57%.

Qasem and Badawi (Qasem Abu Al-Haija, 2022) proposed a CNN based intrusion detection approach for Unmanned Aerial Vehicles (UAVs). Their approach considers encrypted Wi-Fi traffic of three different types of UAVs. They employed a dataset called UAV-IDS-2020 that contains several UAV attacks. They reported best detection accuracy of 99.50% for the binary classifier model.

6.2.1.5. Recurrent neural network. Roy and Cheung proposed in (Roy and Cheung, 2018) DL based technique for detecting attacks within an IoT network using Bi-directional long short-term memory (LSTM). Bidirectional processing utilizes two distinct hidden layers to process input sequences in both forward and backward orientations. They used the UNSW-NB15 dataset to train their model (Moustafa and Slay, 2015). The achieved attack detection accuracy exceeded 95%.

Liang and Znati proposed a DL based DDoS detection scheme (Liang and Znati, 2019). Its input is only packet header information, and it does

not require feature engineering. Their model is based on LSTM. For model learning, Canadian Institute of cyber security dataset CICNIDS-2017 was used (Panigrahi and Borah, 2018). They reported that their model performed nearly the traditional ML methods on the testing case. Their scheme achieved 99.95% accuracy. Their scheme avoided manual feature engineering in order to overcome the significant shortfalls of classic ML techniques. They claimed that their LSTM-based approach could perform better without the use of flow-level statistical characteristics.

Hwang et al. (2019) proposed a DL based mechanism for packet-level semantic classification to identify malicious traffic in the IoT. Their mechanism adopted word embedding to extract semantic meanings from packets. The temporal relationship between the fields in the packet header was discovered using LSTM. ISCX 2012 (Ghurab et al., 2021), USTC-TFC 2016 (Wang et al., 2017), and the Robert Gordon University IoT RGU-IoT dataset were used in the evaluation.

Singh et al. (2021) designed an Edge-Detect NIDS model based on LSTM and GRU for DDoS detection in the IoT. The model was evaluated using the UNSW-NB15 dataset (Moustafa and Slay, 2015). They reported a high testing accuracy of 99%.

Ge et al. (2021) used the simplest form of a feed-forward neural network model (FNN) for the purposes of generality and simplicity. Two classification schemes are considered: binary and multi-class classification, namely mFNN and bFNN. They used embedding to create a dense representation of ports and HTTP request methods in the deep neural network model. Three thick layers and two embedding layers make up the multiclass classifier. The first two dense layers contain 512 neurons each. The ReLU activation function in the first two dense layers was applied. In the last dense layer of four neurons representing four classes, the Softmax function was employed to transform numbers to probabilities. They evaluated their strategy using the Bot-IoT dataset created by Koroniotis et al. (2019). While the accuracy of the binary classifier was greater at 99.99%, the performance of the multi-class classification was reported to be 99.74%. The Adam optimizer method was used to optimize the adaptive learning rate. When labels are mutually exclusive, the loss function to be employed is sparse categorical cross-entropy.

Song et al. proposed an NIDS approach based on a feature selection-conditional Wasserstein generative adversarial network (FCWGAN) and bidirectional LSTM (BiLSTM) (Ma et al., 2022). For feature selection, they used the XGB algorithm with Spearman's correlation coefficient. A small number of samples in the dataset were generated using a conditional Wasserstein generative adversarial network (CWGAN). BiLSTM was applied to complete the training and classification. They used the NSL-KDD (NSL-KDD dataset) and UNSW-NB15 (Moustafa and Slay, 2015) datasets for evaluation and reported accuracy rates of 99.57% and 85.59%, respectively.

Brun et al. (2018) presented a DL-based approach using dense random neural networks (RNN) for intrusion detection in the IoT connected home environment. A test bed of just three devices was used to evaluate the system under simple assaults such as UDP flood, TCP SYN, sleep deprivation attack, barrage attack, and broadcast attack.

Almiani et al. presented a multi-layered RNN for Fog computing security and IoT devices (Almiani et al., 2019). They were able to attain a detection accuracy rate of 92.18% using the NSL-KDD dataset (NSL-KDD dataset). Their model was said to be capable of identifying several assault types, including Probe, Remote to Loca (R2L), and User to Root (U2R).

Qureshi et al. (Q. B et al., 2019) developed an RNN based heuristic intrusion detection system for the IoT. Using the NSL-KDD dataset, they were able to achieve a detection accuracy rate of 92.18% (NSL-KDD dataset). They used two different approaches: the first reduced features to 29 input features, and the second made use of all 41 features in the NSL-KDD dataset (NSL-KDD dataset). 21 hidden layer neurons were set up for both techniques. The acquired accuracy of the case with all features was higher than that of the case with only the reduced method features, at 95.25% and 85.5%.

6.2.1.6. *Deep neural network.* Diro and Chilamkurti deployed a distributed deep learning intrusion detection technique for the IoT/fog networks (Diro and Chilamkurti, 2017). They acquired 123 input features by using the NSL-KDD dataset (NSL-KDD dataset). There are 150 first layer neurons, 120 s layer neurons, 50 third layer neurons, and as many neurons in the final Softmax layer as there are classes. At 50 epochs, their model has different batch sizes. They trained it with dropout in order to prevent overheating. Their DL model outperformed the conventional shallow model's 95.22% accuracy by achieving 99.2% accuracy.

Azmoodeh et al. (2018) used a deep Eigenspace learning approach for malware detection in an Internet of Battlefield Things (IoBT) environment for the military. The device's operational code was transformed into a vector space. They produced a fresh dataset with 128 malicious traffic samples and 1,078 benign traffic samples. The samples were gathered using the VirusTotal3 platform using a secret API between February 2015 and January 2017. They had a 98% accuracy rate using their method.

A fog computing-based technique for IoT intrusion detection utilizing an online sequential extreme learning machine was proposed by Prabavathy et al. ELM is a quick learning technique for feed forward networks with a single hidden layer (Prabavathy et al., 2018). It uses straightforward matrix computations to calculate output weights by selecting input weights and biases at random. A networked, adaptable, and scalable intrusion detection system is made possible by fog computing. They reported faster detection performance at fog nodes by 25% compared to cloud-based solutions using the NSL-KDD dataset (NSL-KDD dataset) for model validation. Their method produced a binary classification accuracy of 97.36% overall.

Ge et al. presented a DL based scheme for detecting IoT intrusions (Ge et al., 2019). For binary and multi-class classification, the model uses feed forward neural networks. It tackles reconnaissance, DoS, DDoS, and information theft. They used Bot-IoT dataset produced by Koroniotis (Koroniotis et al., 2019). They reported detection accuracy for DDoS/DoS close to 99% and 98% for the others.

For intrusion detection in IoT-enabled smart cities, Li et al. suggested a deep migration learning model architecture (Li et al., 2019). KDD CUP 99 (Al Tobi and Duncan, 2018) was selected for model training and evaluation. Although their system had strong temporal complexity detection performance, the detection rates for the four reported attacks were low: DoS 87.19%, U2R 79.52%, R2L 80.83%, and Probe 83.20%.

Chawla and Thamilarasu proposed a DL based framework to identify attacks in the IoT (Chawla and Thamilarasu, 2018). Their framework comprises of an input layer with 56 nodes for a total of 56 received features, three hidden levels, and an output layer. No implementation or testing was reported. While they reported in (Thamilarasu and Chawla, 2019) an average precision rate of 95% and recall rate of 97% for various attack scenarios.

Hussain and Rani (2020) proposed an NIDS for protecting vehicles ad hoc networks. They use a DNN-based strategy that has two main stages: preparation, and detection. They analyzed the approach in terms of average energy consumption, average end to end delay, packet delivery ratio, and average network throughput. Their paper does not report any detection accuracy figures.

Pecori et al. presented a DL based approach for intrusion detection in the IoT (Pecori et al., 2020). Their best accuracy was achieved using 7 layers and 50 epochs for binary classification and 6 layers for multi-class classification. They built an integrated dataset using four different datasets. Due to the limitations of the literature datasets, they combined four different IoT datasets. For both binary and multi-class classification, the reported results were above 0.99 for accuracy, F1 score, and recall. However, such relative big number of hidden layers makes their approach complex and inapplicable for IoT resources constrained environments.

6.2.1.7. General regression neural network. Putschala proposed an NIDS approach for the IoT through mapping the attack types at each layer of the TCP/IP architecture (Putschala, 2017). The KDD Cup 99 dataset (Al Tobi and Duncan, 2018) was used to evaluate his approach. He used GRU and achieved 98.91% detection accuracy.

6.2.1.8. Restricted Boltzmann machine. Dawoud et al. proposed a DL based approach for detecting intrusions in SDN-based IoT (Dawoud et al., 2018). The detection module used the Restricted Boltzmann Machine (RBM) as the intrusion detection engine. For model evaluation, they utilized the KDD Cup 99 dataset (Al Tobi and Duncan, 2018) and achieved precision and accuracy higher than 94%.

6.2.2. Semi supervised deep learning

6.2.2.1. Auto encoder. A conditional variation auto encoder (CVAE) based intrusion detection method was proposed by Lopez-Martin et al. (2017). CVAE is a deep generative model that can provides semi supervised learning through generating outputs based on inputs and auxiliary covariates. Their approach uses CVAE to integrate the intrusion labels inside the decoder layers. To conserve computational resources, their method just requires one training step and may do data recovery (feature reconstruction). They made use of the KDD99-derived NSL-KDD dataset (NSL-KDD dataset), which is a derivation of KDD99 (Al Tobi and Duncan, 2018). They claimed that adding more layers to the encoder or decoder than two layers made training more challenging and did not improve the outcomes. They reported that employing decoder layers one, two, and three led to respective accuracy gains of 77.91%, 80.1%, and 75.47%.

Marta et al. (Catillo et al., 2023) proposed a botnet intrusion detection method based on semi-supervised approach. It is based on an all-in-one AE. They employed N-BaIoT dataset for evaluation after splitting it into several datasets based on the IoT device. The reported results are within the range of 0.9994–0.9997 recall, 0.9999–1.0 precision, 0.0–0.0071 false positive rate and 0.9996–0.9998 F1 score, depending on the device.

6.2.3. Unsupervised deep learning

Mirsky et al. presented Kitsune, which is a neural network based NIDS (Mirsky et al., 2018). It consists of an ensemble of auto encoders called Kitsune Network (KitNET). Their model showed smooth running on simple Raspberry PI. KitNET has two layers: ensemble and output. The ensemble layer is in charge of measuring independent abnormalities. It is a sequence of k three-layer AE. During training, AEs learn normal behavior. During training and execution modes, each AE reports its root mean square error (RMSE) to the output layer. The output layer consists of a three-layer AE. It learns the normal RMSEs of the first layer. This layer is responsible for producing the final anomaly score. This score considers the relationship between subspace abnormalities and naturally occurring noise in the network traffic. They created a simulated dataset that contains malicious traffic of OS Scan, Video Injection, Fuzzing, Active Wiretap, SSDP Flood, ARP MitM, SYN DoS, Mirai, and Regeneration. Online anomaly detection is through an unsupervised ANN called KitNET. The online evaluation results showed good detection performance compared with the other offline algorithms.

Morgera et al. (Amouri et al., 2018) proposed a detection scheme for mobile IoT using a DT based classifier. The core of their scheme relies on the variance of the two traffic types: benign and malicious. The variance is expected to be greater for larger normal traffic (Amouri et al., 2020). For fitting purposes, linear regression is performed to determine a suitable threshold through the slopes of the fitted lines. They used a simulated network for evaluation.

Bhatia et al. presented an unsupervised behavioral learning based classifier using deep learning auto encoders (Bhatia et al., 2019). By using IP spoofing, their method seeks to locate compromised sources and

identify various DDoS attacks. In order to differentiate between known and unknown attack traffic, it took into account the reconstruction error of the trained ML model. They made use of an openly accessible dataset of IoT traffic (Hamza et al., 2019). For identifying slow HTTP DDoS and SYN floods, their method had accuracy close to 100% and was very latency-efficient.

Yin et al. (2020) used LSTM-based recurrent auto encoder and CNN to detect intrusions in IoT. Their approach achieved detection accuracy of 99.62%. Due to the limitations of IoT hardware, the authors reported that further model optimization can be applied to the parameters and architecture.

6.3. Evolutionary computing and rule based techniques

6.3.1. Genetic algorithm

Li and Zhang (2019) also presented a deep belief based model for detecting intrusions in the IoT; but aided with a genetic algorithm (GA). GA iteratively optimizes the number of hidden layers. Neurons in a hidden layer can be adaptively generated. Such a way can significantly reduce the network's complexity. They utilized the KDD Cup 99 dataset (Al Tobi and Duncan, 2018) for model learning and evaluation. Their approach achieved an intrusion detection accuracy of 99%.

6.3.2. Fuzzy logic

Modiri et al. adopted fuzzy logic (FL) for detecting and categorizing intrusions in the IoT (Dovom et al., 2019). It consists of a top-down fuzzy induction structure. The structure's leaves are related by fuzzy predicates applied to input features, whereas the interior nodes of the structure are FL arithmetic operators. They employed the IoT and Vx-Heaven datasets, and their respective detection accuracies were 99.834% and 100%. From their point of view, FL has additional potential for pattern recognition due to its capability of dealing with ambiguities and human brain simulation.

6.4. Combined approaches (Hybrid)

6.4.1. Supervised hybrid techniques

Moustafa et al. adopted an ensemble technique for detecting IoT intrusions (Moustafa et al., 2018). Their technique is particularly suited for botnet attacks against HTTP, MQTT, and DNS protocols. Their technique is based on DT, NB and ANN. Two datasets, namely UNSW-NB15 and NIMS were used. Their model achieved satisfactory overall performance. The detection accuracy for the binary classification ranged between 98.54% and 98.97% for UNSW-NB15 and between 98.29% and 98.36% for NIMS. The individual accuracy of the ANN did not exceed 96.27%.

Liang et al. proposed a hybrid and multi-agent approach (Liang et al., 2019). It uses Blockchain and deep neural network (DNN) to record all communications actions taken by IoT agents. They used the NSL-KDD dataset (NSL-KDD dataset) for model training and evaluation. They reported comparative detection accuracy for DNN of 98% relative to DT based approaches.

Khraisat et al. proposed an ensemble hybrid intrusion detection system (Khraisat et al., 2019). It combines DT and SVM and contains three stages. The first stage contains one C5 DT classifier. The second stage contains one class of SVM. The third stage combines the first two stages using the stacking ensemble method. Their approach was evaluated using the Bot-IoT dataset (Koroniotis et al., 2019) and its reported accuracy was 99.97%.

Saharkhizan et al. designed an approach for detecting attacks in the IoT using advanced DL (Mahdis Saharkhizan et al., 2020). Their approach consists of an ensemble of LSTM classifiers. They used a Modbus network traffic for model evaluation and reported an accuracy of 99%.

Ramadan and Yadav proposed in (Ramadan and Yadav, 2020) a hybrid NIDS system for the IoT based on two stages. The first stage is a

preprocessing phase for feature selection. The enhanced shuffled Frog Leaping algorithm is employed to select the important features. The second stage gets the selected features to be classified using Light CNN with RNN. For evaluation, the NSL-KDD dataset (NSL-KDD dataset) was used and the reported detection accuracy was 91%. Roopak et al. proposed a hybrid NIDS for detecting DDoS attacks using CNN and LSTM. They evaluated their approach using CISNIDS 2017 dataset. The achieved accuracy was reported in (Roopak et al., 2020) as 99.03% and in (Roopak et al., 2019) as 97.16%. Choo et al. presented a hybrid IoT micro-security approach (De La et al., 2019). It includes two main stages. The first is hosted on the client side and employs the CNN model for detecting URL based attacks. The second is hosted at the back-end servers and employs an LSTM model for detecting botnet attacks. The first stage of CNN achieved an accuracy of 94.3%, while the second stage of LSTM achieved an accuracy of 94.80%.

Transfer learning is defined as utilizing the learning generated by a certain model for one activity as the starting point of another model for another objective. Imtiaz and Qusay (Ullah and MAHMOUD, 2021) criticized the approach of Ge et al. (2021) and claimed that it failed to produce precise multiclass classification. So, they adopted the transfer learning concept through CNN instead of FNN. The multiclass classifier is the CNN model, which categorizes 15 different attacks. Their approach was validated using the Bot-IoT (Koroniotis et al., 2019), IoT Network Intrusion, MQTT-IoT-NIDS2020 (H. B et al., 2021), and IoT-23 intrusion detection datasets. The detection rates for Benign, Scan, DoS, and Theft were 99.90%, 99.91%, 99.96%, and 98.10%, respectively.

Priya et al. developed a two-phase hybrid model for detecting intrusions in industrial IoT (Priya et al., 2021). The first phase uses an ensemble blending technique of SVM and NB. RF technique was used to predict class labels. An Adam optimizer based ANN classifier was further used to achieve better prediction accuracy. The second phase is fed by ANN and RF results with considering the highest accuracy value. They tested their model using the WUSTL_IOT 2018 dataset (Aldhaheri et al., 2020), the Bot_IoT dataset (Koroniotis et al., 2019), and the N_BaIoT dataset (Meidan et al., 2018). The highest accuracy reported was 99%.

6.4.2. Unsupervised hybrid techniques

Network intrusions detection is considered a classification problem. Clustering techniques can only produce similar data without class judgment. That is why a classification technique is needed in the case of applying clustering techniques for network intrusion detection problems. There are trials of adopting statistical techniques for filling this gap. For example, Li et al. proposed an AI based approach for a software-defined IoT architecture (Li et al., 2018). It has two phases, feature selection and flow classification. The Bat algorithm is adopted for selecting the features by splitting them into subgroups using the K-means method. Differential evolution was also employed to increase the diversity of individuals. Bat algorithm was inspired by the echolocation behavior of micro bats (Enache and Sg̃, 2015). To achieve great performance, it varies pulse rates of loudness and emission. For the flow classification stage, they optimized RF through weight adjustment of each sample after constructing each tree. The weighted voting mechanism determines the final decision. They used a subset of KDD Cup 99 (Al Tobi and Duncan, 2018) to perform five-class flow classification. Their approach achieved a detection accuracy of 96.03%.

Deng et al. combined the fuzzy c-means clustering (FCM) algorithm and principal component analysis (PCA) (Deng et al., 2018). They selected the intrusion detection evaluation plan of MIT Lincoln laboratory (DARPA) (Anjum et al., 2021) to collect the original traffic. Their approach achieved a detection accuracy of 96.8%. Sheikhan and Bostani presented a hybrid approach for detecting insider and cyber-attacks (Bostani and Sheikhan, 2016a). It is based on an unsupervised ML method and a specification-based method. For distributed detection, their proposed architecture is based on the MapReduce technique (Bostani and Sheikhan, 2016). It consists of intrusion detection agents that identify anomalies and misuse. For intrusion detection, these agents

employ supervised and unsupervised optimum-path forest clustering (OPFC). They used the NSL-KDD dataset (NSL-KDD dataset) and achieved 96.20%.

Other directions adopted DT algorithm for filling this gap. For example, Shukla combined K-means and DT for detecting wormhole attacks (Shukla, 2017). The evaluation was based on a simulation and not tested against other attacks. K-means approach achieved 70–93%, while DT achieved 71–80%; but the combined approach achieved 71–75%. Lee et al. built an abnormal behavior profile for IoT devices (Lee, 2017). To train the classifier to recognize anomalous behavior, they used the dataset produced through profiling. Out of four available data points from a single sensor, they employed two distinct techniques to detect one observed data modification: SVM and k-means clustering with 10 clusters. They reported detection accuracy drops in both algorithms. The accuracy of k-Means algorithm was 92% while 69.5% for SVM. Macdermott et al. proposed a multi-agent approach to threat detection in Healthcare IoT (H-IoT) (Macdermott et al., 2019).

They selected a lightweight multi-agent strategy rather than implementing security, because H-IoT devices have limited resources and limited sensing capabilities. Their approach was reported to use DL; but no details or implementations were reported. For IoT traffic analysis, they tried k-means clustering, but came to the conclusion that DL would provide better analysis and allow for more informed decision-making. As there were no clean distinctions between attack and benign classes, using the clustering algorithm returned a high error rate.

6.5. Unreported techniques

Vishwakarma and Kumar (Vishwakarma, 2019) presented a proposal for adopting honeypot with ML for DDoS detection in the IoT. They were inspired by the ThingPot (Wang et al., 2018). ThinkPot is a virtual IoT honeypot and can intercept multiple botnet malware through simulating various IoT communication protocols and characteristics. ML part is for binary classifying into benign or malicious traffic. The authors did not report any implementation or practical results.

Table 1 presents a comparative analysis for the previous related work in tabular form. It lists the surveyed papers ordered chronologically. The papers' characteristics are listed in terms of the AI Category of used technique, type of learning, learning algorithm, year of publishing, evaluation dataset, reported accuracy, and objective.

7. Discussion and future research opportunities

After discussing previous efforts and contributions of adopting AI algorithms in network intrusion detection in the IoT, this section discusses the research gap and exploring the future research possible opportunities. However, while the literature review demonstrates the potential of AI techniques in detecting network intrusions in IoT, it also shows that further potential is still unexplored. This section analyzes the literature presented in Section 6 to explore the insights behind it.

7.1. Current research directions

This section analyzes all the reviewed literature to reveal the current research directions and explore the possible research possibilities in the future. Fig. 3 shows a Sunburst chart containing the distribution of all the reviewed literature. It is organized starting from the center by categories each with its own percentage, followed by subcategories of type of learning, and finally, the adopted algorithms with their respective percentage in the outer circular layer.

Due to the limited resources of the IoT environment discussed in Section 5.1, a number of studies reported limitations of deploying NIDS on the IoT. Such approaches prefer ML techniques than DL due to its relative lower needed resources. To the contrary, by analyzing the reviewed literature, Fig. 3 shows a slight higher share of studies for adopting ML and DL with 43% versus 40% respectively.

Table 1
AI based NIDS for IoT.

Serial	AI Category	Type of Learning	Learning Algorithm	Main Author	Study	Year	Dataset	Accuracy %	Objective
1	Machine Learning	Supervised	NB	Guezzaz	Guezzaz et al. (2022)	2022	KDD CUP 99	96.45	Snort NIDS and NB for IoT
2			SVM	Al-Yaseen	Al-yaseen et al. (2016)	2016	KDD CUP 99	95.75	Multi-level hybrid NIDS for IoT
3			SVM	Bakker	Bakker et al. (2018)	2018	ISCX	93.00	Detect DDoS attacks in IoT
4			SVM	Koroniotis	Koroniotis et al. (2019)	2019	Bot-IoT	99.98	NIDS for forensics purposes of IoT
5			c-SVM	Ioannou	Ioannou and Vassiliou (2019)	2019	KDD CUP 99	81.00	Detect abnormalities in IoT
6			SVM	Jan	Jan et al. (2019)	2019	CICIDS2017	98.03	Detect unnecessary data injection into IoT
7			SVM	Goyal	Goyal et al. (2019)	2019	Simulated	99.23	Detecting botnets in IoT
8			SVM	Bagaa	Bagaa et al. (2020)	2020	Real sensor data	99.71	NIDS for IoT
9			OC-SVM	Jingyu	Liu et al. (2021)	2021	UNSW-NB15	86.68	NIDS for IoT
10			kNN	Kumar	Kumar and Lim (2019)	2019	Simulated	94.44	Detect IoT malware activity
11			CF-kNN	Pajouh	Pajouh et al. (2019)	2019	NSL-KDD	84.86	Layered intrusion detection for IoT
12			kNN	Alsamiri	Alsamiri and Alsubhi (2019)	2019	Bot-IoT	99.00	NIDS for IoT
13			DT	Bahsi	Bahsi et al. (2018)	2018	Simulated	98.97	IoT botnet detection
14			DT	Aloqaily	Aloqaily et al. (2019)	2019	Simulated	99.43	
15			DT (J48)	Anthi	Anthi et al. (2019)	2019	Synthetic	98.00	
16			DT (BE)	Illy Manimurugan	Illy et al. (2019)	2019	NSL-KDD	85.81	Securing Fog-to-Things
17			DT		Manimurugan et al. (2020)	2020	CICNIDS 2017	98.37	
18			RF	Doshi	Doshi et al. (2017)	2017	Simulated	99.80	Intrusion detection in smart medical IoT
19			RF	Dwyer	Dwyer et al. (2018)	2018	Real Dataset	99.00	DDoS detection in IoT
20			RF	Chaudhary	Chaudhary and Gupta (2019)	2019	Simulated	99.17	Profiling IoT botnet traffic using DNS
21			RF	Hasan	Hasan et al. (2019)	2019	Pahl	99.40	DDoS detection in IoT
22			RF	Alrashdi	Alrashdi et al. (2019)	2019	UNSW-NB15	99.34	Intrusion detection and classifying in IoT
23			RF	Thamilarasu	Thamilarasu et al. (2020)	2020	Simulated	100.0	NIDS for IoT
24			RF	Hammoudeh	Hammoudeh and Aljaberi (2021)	2021	KDDCup99	89.39	Intrusion detection for medical IoT
25			RF	Eskandari	Eskandari et al. (2020)	2020	Simulated	99.00	NIDS for IoT
26			RF	Saied	Saied and Guirguis (2023)	2023	N-BaIoT	99.99	NIDS for IoT
27			XGB	Kumar	Kumar et al. (2019)	2019	Synthetic	99.88	Peer-to-Peer Bot-net Detection
28			XGB	Liu	Liu et al. (2020)	2020	NSL-KDD	97.00	NIDS for Contiki-NG-Based IoT
29			XGB	Alqahtani	Alqahtani et al. (2020)	2020	N-BaIoT	99.96	IoT Botnet Attack Detection
30			XGB	Khan	Khan et al. (2022)	2022	Elnour et al. HVAC dataset	99.98	
31			XGB	Ashraf	Ashraf et al. (2022)	2022	CIC-IDS2018, N-BaIoT, KDD Cup 99	98.96	Attack detection for HVAC
32			XGB/DT	Alissa	Alissa et al. (2022)	2022	UNSW-NB15	94.00	NIDS for Blockchain enabled IoT Healthcare Applications
33			XGB	Garg	Garg et al. (2022)	2022	BoT-IoT, IoT-23, CIC-DDoS-19	94.49	Botnet attack detection in IoT
34			XGB	Awotunde	Awotunde et al. (2023)	2023	ToN-IoT	99.73	Attacks Identification: IoT attacks and DDoS attacks
35			XGB	Rani	Rani et al. (2023)	2023	DS2OS	99.92	NIDS for Smart home IoT
36			LGB	Garg	Garg et al. (2022)	2022	BoT-IoT, IoT-23, CIC-DDoS-19	94.79	Attacks Identification: IoT attacks and DDoS attacks
37			LGB	Bhoi	B. B et al. (2022)	2022	IoT dataset	99.99	Identification of Malicious Access in the IoT Network
38			PSO-LGB	Bhoi	B. B et al. (2022)	2022	IoT dataset	100.0	Anomaly detection in IoT
39			GSA-LGB	Bhoi	B. B et al. (2022)	2022	IoT dataset	100.0	
40			ADB	Dash	Dash and Rao (2020)	2020	DS2OS	95.00	

(continued on next page)

Table 1 (continued)

Serial	AI Category	Type of Learning	Learning Algorithm	Main Author	Study	Year	Dataset	Accuracy %	Objective
41	Deep Learning	Supervised	ADB	Krishna	Krishna and Thangavelu (2021)	2021	NSL-KDD and NBaIoT	99.30	Attack detection in IoT
42			ADB	Qasem	Al-Haija et al. (2021)	2021	TON_IoT	99.70	NIDS for IoT
43			EL ADB	Hazman	Hazman et al. (2022)	2022	IoT-23, Bot-IoT, Edge-IIoT	99.90	NIDS for Smart cities IoT
44			ADB	Al-Haija	Al-haija and Al-Dala'ien (2022)	2022	N-BaIoT	97.30	Botnet attack detection in IoT
45			RUS					97.70	
46			ELBA	Hodo	Hodo et al. (2016)	2016	Simulated	99.60	DDoS detection in IoT
47			MLP					99.40	
48			ANN	Cañedo	Cañedo and Skjellum (2016)	2016	Simulated	99.00	NIDS for the IoT
49			ANN	Chang	Chang and Hsieh (2018)	2018	DARPA with CAIDA	99.80	DDoS detection in IoT
50			ANN	Mohamed	Mohamed et al. (2018)	2018	UNSW -NB15	89.00	NIDS for IoT
51			ANN	Soe	Soe et al. (2019a)	2019	Bot-IoT	100.0	DDoS detection in IoT
52			ANN	Fatayer	Fatayer and Azara (2019)	2019	KDD CUP 99	99.86	NIDS for the IoT
53			ANN	Khatun	Khatun et al. (2019)	2019	Simulated	77.51	Malicious Nodes Detection in IoT
Serial	Problem Category	Type of Learning	Learning Algorithm	Main Author	Study	Year	Dataset	Accuracy %	Objective
54	Deep Learning	Supervised	ANN	Lakshmi	Lakshmi and Prashanth (2022)	2022	Simulated	NA	Black hole attack detection and prevention in HWSN
55			ANN	Soe	Soe et al. (2019b)	2019	N_BaIoT	100.0	NIDS for IoT
56			AE	Meidan	Meidan et al. (2018)	2018	N-BaIoT	100.0	NIDS for IoT
57			APSO-CNN	Kan	Kan et al. (2021)	2021	N-BaIoT	96.00	NIDS for IoT
58			CNN	Idrissi	Idrissi et al. (2021)	2021	Bot-IoT	99.94	Botnet attacks detection in IoT
59			CNN	Fatani	Fatani et al. (2021)	2021	KDD CUP 99, NSL-KDD, Bot-IoT, CICNIDS2017	99.63	Feature extraction
60			CNN	Zhou	Jingjing et al. (2022)	2022	WSN-DS-2016	99.57	NIDS for WSN
61			CNN	Qasem	Qasem Abu Al-Haija (2022)	2022	UAV-IDS-2020	99.50	NIDS for networked UAVs
62			LSTM	Roy	Roy and Cheung (2018)	2018	UNSW-NB15	95.00	NIDS for IoT
63			LSTM	Liang	Liang and Znati (2019)	2019	CICNIDS-2017	99.95	DDoS detection in IoT
64			LSTM	Hwang	Hwang et al. (2019)	2019	ISCX2012	99.99	Classifying Malicious Traffic at the Packet Level
65			LSTM Ensemble of LSTM	Sing	Singh et al. (2021)	2021	USTC-TFC2016	99.99	DDoS detection in IoT
66							RGU-IoT	100.0	
67							UNSW-NB15	99.00	
68			BiLSTM	Ge	Ge et al. (2021)	2021	Bot-IoT	99.99	NIDS for IoT
69			RNN	Song	Ma et al. (2022)	2022	NSL-KDD	99.57	NIDS for IoT
70							UNSW-NB15	85.59	
71			RNN	Brun	Brun et al. (2018)	2018	Simulated	NA	NIDS for connected home Fog computing security and IoT devices
72			RNN	Almiani	Almiani et al. (2019)	2019	NSL-KDD	92.18	
73			RNN	Qureshi	Q. B et al. (2019)	2019	NSL-KDD	95.25	Heuristic intrusion detection for IoT
74			DNN	Diro	Diro and Chilamkurti (2017)	2017	NSL-KDD	99.20	Fog computing based NIDS in the IoT
75			DNN	Azmoodeh	Azmoodeh et al. (2018)	2018	Simulated	98.00	Malware detection in IoT
76			DNN	Prabavathy	Prabavathy et al. (2018)	2018	NSL-KDD	97.36	Fog computing based NIDS in IoT
77			DNN	Ge	Ge et al. (2019)	2019	Bot-IoT	98.00	NIDS for IoT
78			DNN	Li	Li et al. (2019)	2019	KDD CUP 99	87.19	Intrusion detection in smart cities IoT
79			DNN	Chawla	Chawla and Thamilarasu (2018)	2019	NA	NA	Attacks identification in IoT
80			DNN	Hussain	Thamilarasu and Chawla (2019)	2020	NA	NA	NIDS for protecting vehicles ad hoc networks
					Hussain and Rani (2020)				(continued on next page)

Table 1 (continued)

Serial	AI Category	Type of Learning	Learning Algorithm	Main Author	Study	Year	Dataset	Accuracy %	Objective
81	Problem Category Evolutionary Computing Rule Based	Semi Supervised	DNN	Pecori	Pecori et al. (2020)	2020	Synthetic	99.00	NIDS for the IoT
82			GRNN	Putchala	Putchala (2017)	2017	KDD CUP 99	98.91	NIDS for the IoT
83			RBM	Dawoud	Dawoud et al. (2018)	2018	KDD CUP 99	94.00	Intrusion detection in SDN-based IoT
84			AE	Lopez-Martin	Lopez-Martin et al. (2017)	2017	NSL-KDD	80.10	NIDS for IoT
85			AE	Marta	Catillo et al. (2023)	2023	N-BaIoT	NA	IoT Botnet Detection
86		Unsupervised	DNN + DT	Mirsky	Mirsky et al. (2018)	2018	Simulated	NA	NIDS for IoT
87			AE	Morgera	Amouri et al. (2018)	2018	Simulated	NA	Anomaly detection for mobile IoT
88			AE	Bhatia	Bhatia et al. (2019)	2019	Hamza Dataset	100.0	Anomaly detection for IoT
89			AE	Yin	Yin et al. (2020)	2019	Simulated	99.62	NIDS for IoT
90			GA	Li	Li and Zhang (2019)	2019	KDD CUP 99	99.00	NIDS for IoT
91	Hybrid	Supervised	FL	Modiri	Dovom et al. (2019)	2019	IoT	99.83	Intrusion detection & categorization in the IoT
92		Supervised	DT + NB + ANN	Moustafa	Moustafa et al. (2018)	2018	Vx-Heaven	100.0	NIDS for IoT
93							UNSW-NB15 + NIMS	98.97	
94							NSL-KDD	85.81	
95							NSL-KDD	98.00	
96							Bot-IoT	99.97	
97		Unsupervised	DT + NB + ANN	Saharkhizan	Mahdis Saharkhizan et al. (2020)	2020	Modbus network traffic	99.00	NIDS for IoT
98							NSL-KDD	91.00	
99							CISNIDS2017	97.16	
100							N_BaIoT	94.80	
101							BoT-IoT, IoT-23, MQTT-IoT-NIDS2020	99.96	
102		Unsupervised	CNN + LSTM	Choo	De La et al. (2019)	2020	WUSTL_IIoT- 2018, N_BaIoT, Bot_IoT	99.00	Intrusion detection for IIoT
103			bFNN + mFNN	Imtiaz	Ullah and MAHMOUD (2021)	2021	NSL-KDD	96.20	Detecting insider and cyber-attacks
104			SVM + NB	Priya	Priya1 et al. (2021)	2021	KDD Cup 99	96.03	NIDS for Software Defined IoT
105			K-means + OPFC	Sheikhan	Bostani and Sheikhan (2016a)	2016	DARPA	96.80	NIDS for IoT
106			OPFC + MOPF	Li	Li et al. (2018)	2018	Synthetic	75.00	Wormhole attacks detection in 6LoWPAN IoT
107			FCM + PCA	Deng	Deng et al. (2018)	2018	Simulated	93.70	Threat detection for Healthcare IoT
108			K-means + DT	Shukla	Shukla (2017)	2017	NW dataset	NA	NIDS for H-IoT
109			K-means + DT	Lee	Lee (2017)	2017			
110			K-means + DT	Macdermott	Macdermott et al. (2019)	2019			

Among the ML approaches, the one with the highest adoption is XGB with an overall percentage of 7%. While for DL approaches, the highest adoption is for LSTM with an overall percentage of 9%. There are many algorithms in both ML and DL that have not been addressed, such as Categorical Boosting algorithm (also known as CatBoost or CAB) ([Ibrahim et al., 2020](#)), Histogram-based Gradient Boosting Classification Tree (HGB), Classification and Regression Trees (CART) and Gradient Descent boosting (GDB).

The survey also highlights a modest adoption of evolutionary computing in the field of network intrusion detection. There is only one study on GA adoption. In the field of rule based AI, also, only one paper is reported that presents adopting FL and evaluates against two different datasets. It is clear that such a low number of approaches represents a significant potential for further research to explore its capabilities.

7.2. Using benchmarking datasets

Benchmarking plays an important role in improving research. Referring to [Fig. 4](#), the majority of presented methods were tested using simulated or outdated datasets. A few of them have been tested using well known standard datasets e.g. NSL-KDD ([Almiani et al., 2019](#)), and N-BaIoT dataset ([Naveed et al., 2020](#)). For example, [Chang and Hsieh \(2018\)](#) used the DARPA dataset ([Anjum et al., 2021](#)) with the CAIDA DDoS, which were collected in 1999 and 2007, respectively. Using such outdated datasets may not accurately reflect the current threat landscape of IoT environment and may lead to overly optimistic results.

[Fig. 5](#) shows a word cloud figure for the used datasets. It represents each dataset name according to its related adoption percentage in the discussed literature. However, NSL-KDD is suggested to overcome some problems of the KDD CUP 99, it still may not be a perfect representative

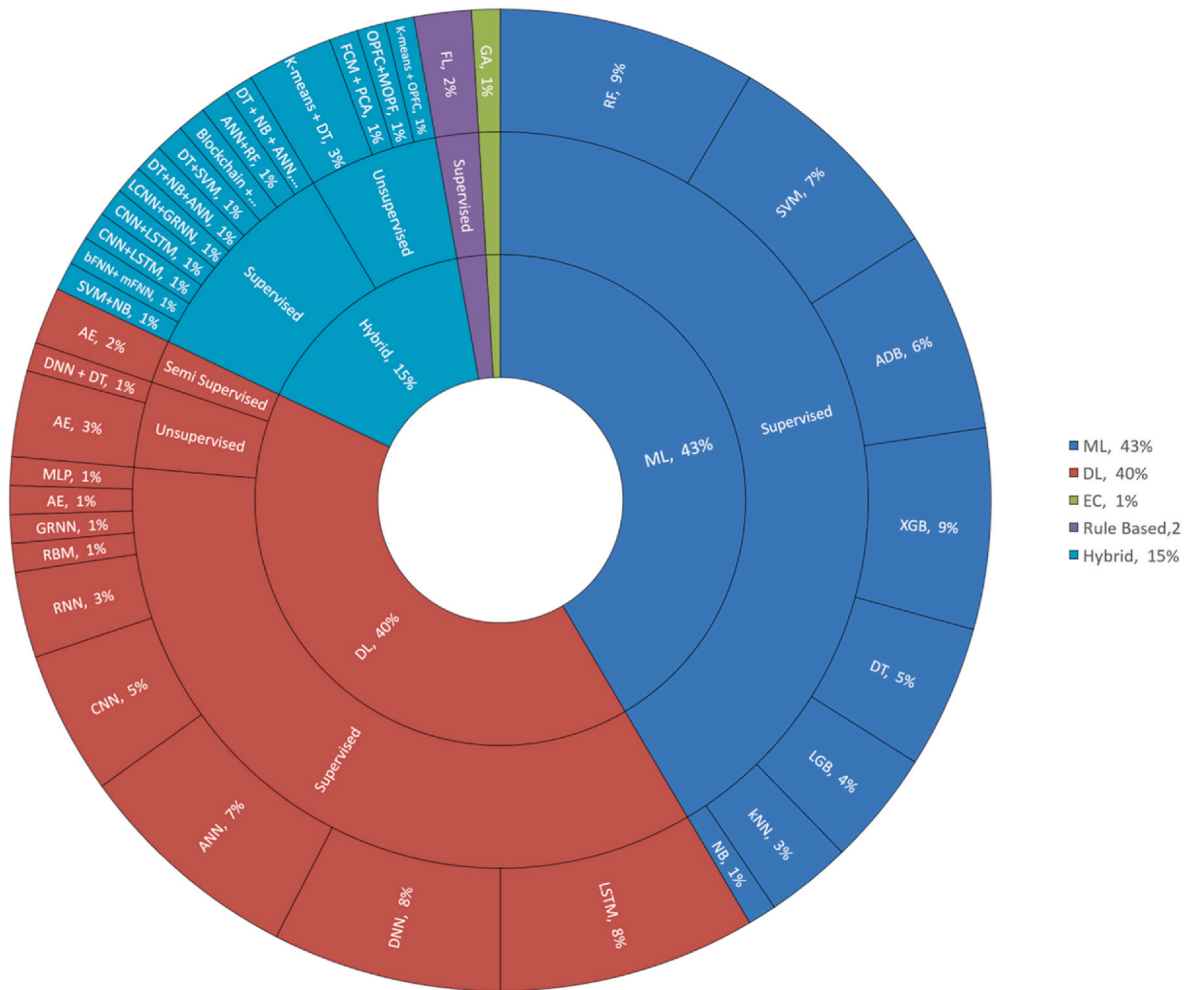


Fig. 3. Current research directions.

of IoT real networks (NSL-KDD dataset). Other approaches use a mixed dataset from some of these datasets. It leads to missing a unique datum for performance evaluation of the proposed approaches. As each of them depends on a different simulated dataset, the reported high accuracy values cannot be considered in benchmarking with other approaches.

On the other side, collecting real-world datasets for evaluating intrusion detection systems in IoT environments is a challenging task due to privacy concerns and the difficulty of accessing real-world IoT systems. One possible solution for such challenges may be to use hybrid datasets that combine real-world and simulated data, developing standardized datasets that are representative of IoT real-world scenarios, and using techniques such as data augmentation to generate synthetic data that can supplement real-world datasets.

7.3. Evidence of reported results

Most of the surveyed studies reported high accuracy rates for their approaches; but no evidence of such values was presented. A few studies presented the implementation of their approaches in shared repositories. Sharing implementation can provide evidence of the reported values while also allowing other researchers to check and improve. Fig. 6 shows the reported accuracy rates for the pre discussed literature in the form of Radar diagrams. Its axis represents the accuracy rates in ascending order from the center to the edge on a scale from 73% to 100%. It demonstrates that there is still room for improvement in detection accuracy.

7.4. Network intrusion detection problem type

Through analyzing the reviewed literature, Fig. 7 highlights the dominance of adopting supervised learning with 90% of the overall surveyed studies. Such a large ratio is mainly due to the nature of intrusion detection as a classification-oriented problem. The other approaches adopting unsupervised techniques are either dependent on adopting statistical techniques or relying on a hybrid approach with a DT algorithm. Such a modest number of approaches to unsupervised learning represents a strong potential for further research in this area, especially, when considering the fact that most of the available datasets are unlabeled.

Reinforcement learning (RL) is not widely used in network intrusion detection in the IoT. It can present a dynamic behavior based approach for using the reward and punishment technique. It has the potential to be a useful approach for intrusion detection in IoT systems. It involves an agent learning to make decisions by interacting with an environment and receiving feedback in the form of rewards or penalties. In the context of IoT intrusion detection, RL can be used to train an agent to make decisions about whether a particular network activity is normal or suspicious based on the received data. It can adapt to changing environments and learn from experience. IoT systems are often dynamic and complex, and traditional intrusion detection approaches may not be able to keep up with the constantly changing threat landscape. RL can help to address this challenge by enabling an agent to learn and adapt in real-time. However, there are also several challenges associated with using RL for intrusion detection in IoT systems. For example, RL requires large

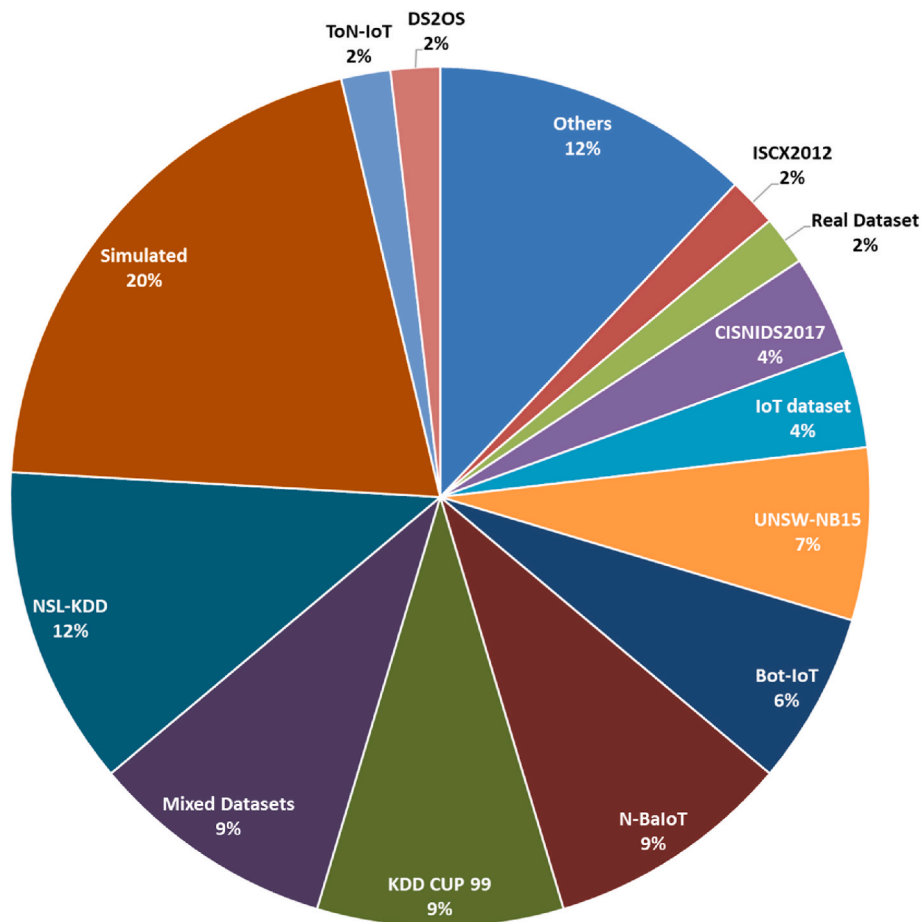


Fig. 4. Research directions distribution of used datasets.



Fig. 5. Word cloud for used datasets.

amounts of data to train the agent, which may be difficult to obtain in the case of IoT systems that have limited resources. Additionally, RL algorithms can be computationally expensive, which may be a concern for resource-constrained IoT devices. Despite these challenges, further research is needed to fully understand the potential of RL for IoT intrusion detection and to develop practical solutions that can be

deployed in real-world settings.

8. Conclusion

AI has attracted attention from both academic literature and business for enhancing the IoT security. The potential of AI in IoT intrusion

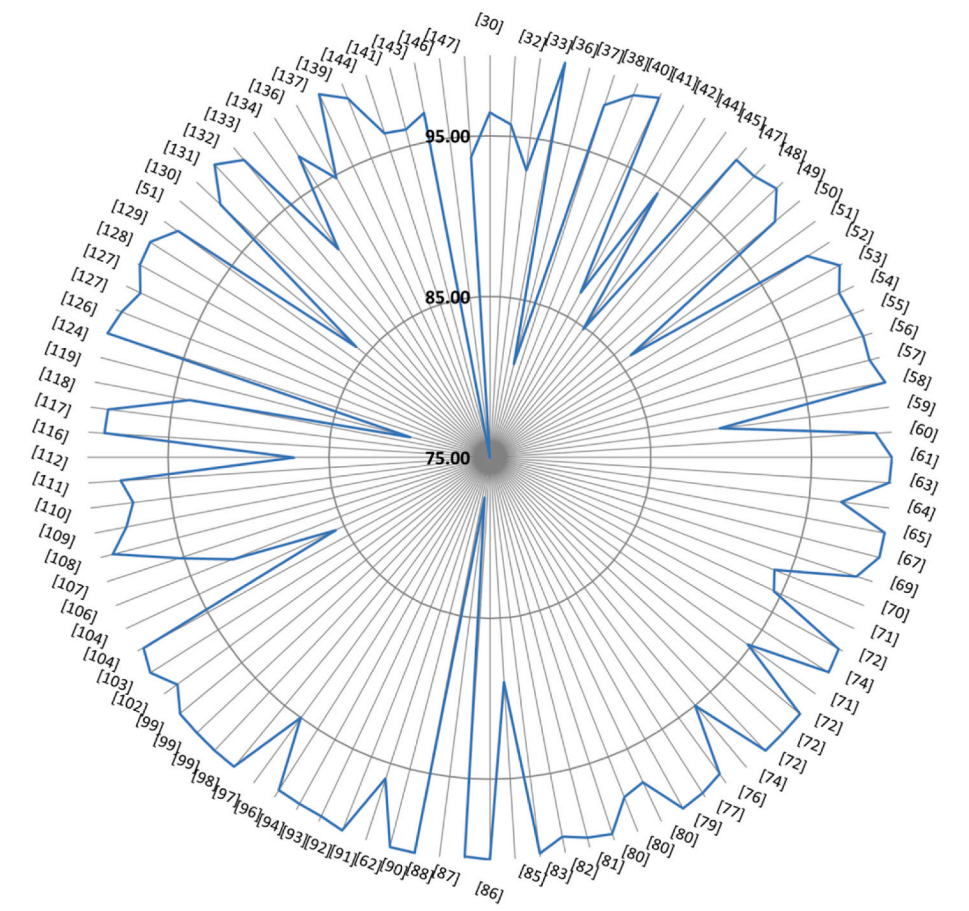


Fig. 6. Reported accuracy rates along publications.

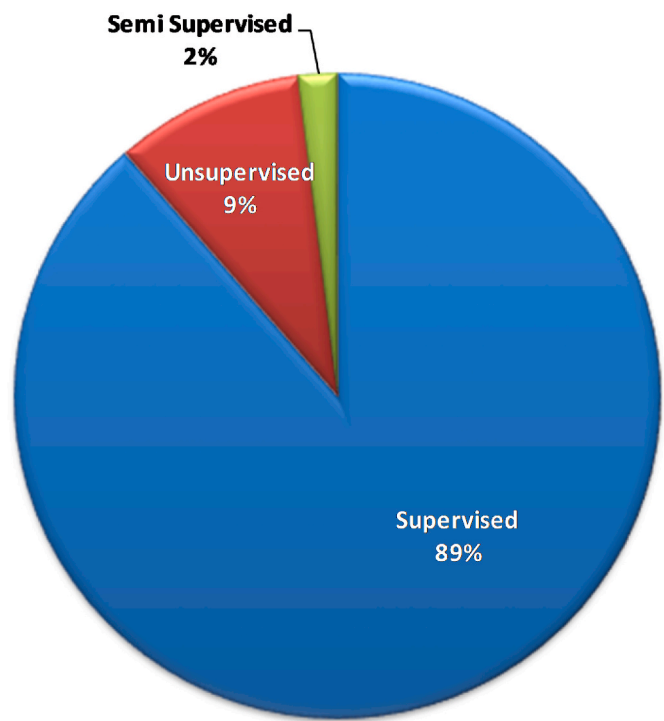


Fig. 7. Research Directions Distribution based on Learning Type.

detection is immense and has been the subject of much research in recent years. AI-based intrusion detection systems have the potential to improve the security of IoT devices and networks by identifying and mitigating threats in real-time. Through a systematic literature review of AI-based network intrusion detection approaches, this paper aims to investigate the potential of AI in enhancing IoT security. The paper has highlighted the various AI approaches that have been proposed for intrusion detection in IoT systems. The literature is categorized based on the adopted AI category and the type of learning including supervised learning, semi-supervised, and unsupervised learning.

The review discussed some of the challenges associated with the use of AI for intrusion detection in IoT systems, such as the need for large and diverse datasets and the challenge of balancing the need for accuracy with the need for efficiency in resource-constrained environments. The analysis revealed that, despite achieving high detection rates, the majority of those studies used simulated or outdated datasets. Relying on such datasets makes the results unrepresentative of the real world and increases the uncertainty of their performance in reality. It showed that despite these challenges, there is significant potential for AI-based intrusion detection systems to enhance the security of IoT systems and protect against emerging threats.

The paper presents some analytics for the reviewed literature for the sake of determining future possible research. It discussed several possible directions of future research. Going forward continued research and development in this area will be crucial to realizing the full potential of AI in IoT intrusion detection and ensuring the security and privacy of IoT devices and networks.

CRediT authorship contribution statement

Mohamed Saied: is the corresponding author, performed the, Conceptualization, determined the research, Methodology, Any disagreement over inclusion or exclusion was discussed with the senior author (S.G.) and a consensus reached, prepared and, Writing – original draft, Writing – review & editing, conducts the, Formal analysis, of the results, designed the. **Shawkat Guirguis:** performed the, Conceptualization, surveyed the related work, determined the research, Methodology, collected the data, independently performed the searches and examined titles and abstracts to exclude irrelevant papers and produce a list of studies for full-text review in an iterative process. Any disagreement over inclusion or exclusion was discussed with the senior author (S.G.) and a consensus reached, prepared and, Writing – original draft, reviewed the reviews and, Writing – review & editing, to avoid incidental plagiarism, conducts the, Formal analysis, of the results, Supervision, the whole research, All authors discussed the results and contributed to the final manuscript. **Magda Madbouly:** performed the, Conceptualization, surveyed the related work, determined the research, Methodology, collected the data, independently performed the searches and examined titles and abstracts to exclude irrelevant papers and produce a list of studies for full-text review in an iterative process. Any disagreement over inclusion or exclusion was discussed with the senior author (S.G.) and a consensus reached, Writing – review & editing, designed the.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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